

25 C

$$\sqrt{\left(2\pi d\right)^{-1}\left(2\pi d\right)^{-1}}=\sqrt{L}\left(2\pi d\right)^{-1}$$

$$\varepsilon=B\left(L/3\right)v$$

$$i=\frac{\varepsilon}{r\left(L/3\right)}=\frac{B\left(L/3\right)v}{r\left(L/3\right)}=\frac{Bv}{r}$$

At constant speed, $F_{\text{ext}} = F_B$

$$F_{\text{ext}}=Bi\left(\frac{L}{3}\right)=B\left(\frac{Bv}{r}\right)\left(\frac{L}{3}\right)=\frac{B^2Lv}{3r}$$

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